Publishing and Sharing in Tableau

Q1. What is the key difference between publishing a dashboard on Tableau Public versus Tableau Server/ Cloud in terms of accessibility and data security?

Answer:

There are majorly two differences between Tableau Public vs Tableau Server/Cloud.

Tableau Public

Accessibility: Dashboards published here are open to everyone on the internet. Anyone with the link can view it, and it may even be discoverable through search engines.
Data Security: There's no privacy or security layer—your data is exposed publicly. Sensitive or confidential datasets should never be published here.

Tableau Server / Tableau Cloud

Accessibility: Dashboards are shared within your organization or with specific users. Access is controlled through permissions and authentication.
Data Security: Provides enterprise-level security with encryption, user roles, and governance. Data remains private and protected, suitable for sensitive business information.

Q2. When publishing a Tableau dashboard, what happens to a live connection compared to an extract? How does this affect data refresh and performance?

Answer:
Key difference between connection and extract are: -

Live Connection

Behaviour: The dashboard stays connected directly to the underlying data source (e.g., SQL database, cloud warehouse).
Data Refresh: Always pulls the latest data whenever a user opens or interacts with the dashboard. No need for manual refreshes.
Performance: Can be slower, especially if the data source is large or the queries are complex, since every interaction sends a query back to the database.

Extract
Behaviour: Tableau takes a snapshot of the data and stores it in a highly optimized format (. hyper file).
Data Refresh: Data is static until you schedule or manually refresh the extract. Users see the data as of the last refresh.
Performance: Usually faster, because Tableau queries the local extract instead of hitting the live database every time. Great for large datasets or when the source is slow.

Q3. In Tableau Server or Cloud (conceptually), which feature allows administrators to control which users can view, interact with, or edit a published dashboard?

Answer:
The feature that allows administrators to control access is permissions and user roles.

Permissions: Admins can set rules for each dashboard or workbook, defining who can view, interact with, or edit it.
User Roles: Roles (like Viewer, Explorer, Creator, or Administrator) determine the level of functionality available to each user across the platform.
Granularity: Permissions can be applied at multiple levels — project, workbook, or even specific dashboard views.

Q4. If you want users to view a dashboard but not download the underlying data, which permission setting would you adjust in Tableau Server?

Answer:

If I want that user to view a dashboard but not download the underlying data, you need to adjust the Download Data permission.

Download Data (Summary Data / Full Data):

>> If enabled, users can export or download the data behind the visualizations.
>> If disabled, users can still view and interact with the dashboard but cannot access or export the raw dataset.

Q5. Name two methods by which a Tableau dashboard can be embedded into a web page or website.

Answer:

Two Methods to Embed a Tableau Dashboard into a Website

1. **Embed Code**

>> Tableau provides an embed code that you can copy and paste directly into your webpage.
>> This method allows the dashboard to be displayed seamlessly inside the site, just like embedding a YouTube video.

2. **JavaScript API**

>> Tableau's JavaScript API gives developers more control.

Q6. List two best practices you should follow when preparing a dashboard for public sharing to ensure data security and privacy.

Answer:

Best Practices for Public Sharing in Tableau

1. **Remove or Mask Sensitive Data**

>> Exclude confidential fields (like customer names, emails, financial details).
>> Aggregate data where possible (e.g., show totals instead of individual records).

2. **Apply Data Anonymization or Filters**

>> Use calculated fields or filters to hide personally identifiable information (PII).
>> Limit the scope of data shown so only relevant, non-sensitive insights are visible.

Q7. After publishing a dashboard, users report a "Data not found" error.
What is the most common cause related to data source settings?

Answer:

Common Cause of "Data not found" Error

The most frequent reason is data source connection settings when publishing:

1. **Live Connection Not Accessible:**

The dashboard may have been built with a live connection to a local database or file that Tableau Server/Cloud cannot reach. For example, if the data source is on your desktop or a private network, the server won't be able to access it.

2. **Extract Not Published Properly:**

If you used an extract but didn't publish it along with the workbook, the server won't have the data file to query.

Q8. When embedding a Tableau dashboard into a secure internal portal, what two additional security measures should be implemented to ensure only authorized users can access the embedded content?

Answer:

Security Measures for Embedding Tableau Dashboards in Internal Portals

When embedding into a secure internal portal, you should implement these two key measures:

1. **Single Sign-On (SSO) / Authentication Integration**

>> Ensure that the portal uses enterprise authentication (e.g., Active Directory, SAML, OAuth).
>> This way, only users who are logged in with valid organizational credentials can access the embedded dashboard.

2. **Row-Level Security (RLS) or Permission Controls**

>> Configure Tableau permissions so that users only see the data they are authorized to view.
>> RLS ensures that even within the same dashboard, different users may see different subsets of data depending on their role.